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Electronic device, control method and control apparatus thereof, and terminal device

Abstract

An electronic device is disclosed. A first driving mechanism is provided in an inner chamber, which is capable of driving a movable module to retract into or extend out of a housing through a through hole. The movable module includes a first bracket, a second driving mechanism, a second bracket rotatably disposed on the first bracket, a functional module including an anti-shake micro-cloud gimbal and a camera movably disposed on the anti-shake micro-cloud gimbal, and a third driving mechanism. The third driving mechanism is disposed on the second bracket, connected to the functional module, and drives the functional module to rotate around a first axis. The second driving mechanism is disposed on the first bracket, and drives the second bracket to rotate around a second axis. The functional module is rotatably disposed on the second bracket, and is capable of rotating with the second bracket.

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Background/Summary

(1) This application is a continuation application of PCT International Application No. PCT/CN2021/139006 filed on Dec. 17, 2021, which claims priority to Chinese Patent Application No. 202011517166.2, filed with the China National Intellectual Property Administration on Dec. 21, 2020 and entitled “ELECTRONIC DEVICE, CONTROL METHOD AND CONTROL APPARATUS THEREOF, AND TERMINAL DEVICE”, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) This application relates to the field of shooting technology for electronic devices, and in particular, to an electronic device, a control method and control apparatus thereof, and a terminal device.

BACKGROUND

(2) With the rapid development of electronic devices, users propose increasingly high requirements for shooting functions of electronic devices. Specifically, users propose increasingly high requirements for clarity of shots, field of view for shooting, and the like.

(3) In the related art, camera modules are all fixedly provided in electronic devices. When a user is taking photos with an electronic device, it is difficult for the user who holds the electronic device by hand to maintain a stationary shooting posture, so the user tends to shake, resulting in blurry images shot by the electronic device. This leads to poor anti-shake effects of the electronic device in the related art.

SUMMARY

(4) According to a first aspect, this application discloses an electronic device including a housing, a first driving mechanism, and a movable module, where the housing has an inner chamber and a through hole communicating with the inner chamber, the first driving mechanism is disposed in the inner chamber, and the first driving mechanism is connected to the movable module and capable of driving the movable module to retract through the through hole into the housing or at least partially extend out of the housing; where the movable module includes a first bracket, a second driving mechanism, a second bracket, a functional module, and a third driving mechanism, where the functional module is rotatably disposed on the second bracket, the third driving mechanism is disposed on the second bracket, the third driving mechanism is connected to the functional module and drives the functional module to rotate around a first axis, the second bracket is rotatably disposed on the first bracket, the second driving mechanism is disposed on the first bracket, the second driving mechanism drives the second bracket to rotate around a second axis, the functional module is capable of rotating with the second bracket, and the first axis and the second axis intersect or lie on different planes; the functional module includes an anti-shake micro-cloud gimbal and a camera, the camera being movably disposed on the anti-shake micro-cloud gimbal; and the electronic device has a first anti-shake state and a second anti-shake state, where in a case that the electronic device is in the first anti-shake state, at least one of the second driving mechanism and the third driving mechanism drives the functional module to rotate; and in a case that the electronic device is in the second anti-shake state, the camera is capable of rotating relative to the anti-shake micro-cloud gimbal.

(5) According to a second aspect, this application discloses a control method of electronic device, where the electronic device is the electronic device according to the first aspect, and the control

method includes: receiving control information; in a case that the control information is first control information, controlling at least one of the second driving mechanism and the third driving mechanism to drive the functional module to rotate; and in a case that the control information is second control information, controlling the camera to rotate relative to the anti-shake micro-cloud gimbal.

(6) According to a third aspect, this application discloses a control apparatus of electronic device, where the electronic device is the electronic device according to the first aspect, and the control apparatus includes: a receiving module configured to receive control information; a first control module configured to, in a case that the control information is first control information, control at least one of the second driving mechanism and the third driving mechanism to drive the functional module to rotate; and a second control module configured to, in a case that the control information is second control information, control the camera to rotate relative to the anti-shake micro-cloud gimbal.

(7) According to a fourth aspect, this application discloses a terminal device including a processor, a memory, and a program or instructions stored in the memory and capable of running on the processor, where when the program or instructions are executed by the processor, the steps of the control method according to the second aspect are implemented.

(8) According to a fifth aspect, this application discloses a readable storage medium, where the readable storage medium stores a program or instructions, and when the program or instructions are executed by a processor, the steps of the control method according to the second aspect are implemented.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) To describe the technical solutions in the embodiments of this application or in the background more clearly, the following briefly describes the accompanying drawings required for describing the embodiments or the background. Apparently, persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 is a schematic structural diagram of an electronic device with a movable module extending out of a housing according to an embodiment of this application;

(3) FIG. 2 is a schematic structural diagram of an electronic device with a movable module retracting into a housing according to an embodiment of this application; and

(4) FIG. 3 is a schematic diagram of a terminal device.

(5) Reference signs in the accompanying drawings are described as follows: **100.** housing; **110.** through hole; **120.** rotating shaft; **200.** first driving mechanism; **300.** movable module; **310.** first bracket; **320.** second driving mechanism; **330.** second bracket; **340.** functional module; **341.** anti-shake micro-cloud gimbal; **342.** camera; **343.** fill light assembly; **350.** third driving mechanism; **400.** base; **500.** elastic structural member; **600.** guide member; **700.** sleeve member; **800.** first circuit board; **900.** second circuit board; **1200.** terminal device; **1201.** radio frequency unit; **1202.** network module; **1203.** audio output unit; **1204.** input unit; **12041.** graphics processing unit; **12042.** microphone; **1205** sensor; **1206.** display unit; **12061** display panel; **1207.** user input unit; **12071.** touch panel; **12072.** other input devices; **1208.** interface unit; **1209.** memory; **1210.** processor; and **1211.** power supply.

DESCRIPTION OF EMBODIMENTS

(6) To make the objectives, technical solutions, and advantages of this application clearer, the following clearly describes the technical solutions of this application with reference to specific embodiments of this application and corresponding accompanying drawings. Obviously, the described embodiments are merely some but not all of the embodiments of this application. All

other embodiments obtained by persons of ordinary skill in the art based on the embodiments of this application without creative efforts fall within the protection scope of this application.

(7) The technical solutions disclosed in the embodiments of this application are described in detail below with reference to the accompanying drawings.

(8) Referring to FIG. 1 and FIG. 2, an embodiment of this application discloses an electronic device. The disclosed electronic device includes a housing **100**, a first driving mechanism **200**, and a movable module **300**.

(9) The housing **100** provides a foundation for mounting the first driving mechanism **200** and the movable module **300** and also provides protection for the first driving mechanism **200** and the movable module **300**. The housing **100** has an inner chamber and a through hole **110** communicating with the inner chamber. The first driving mechanism **200** is disposed in the inner chamber so that the housing **100** can provide protection for the first driving mechanism **200**.

(10) The first driving mechanism **200** is connected to the movable module **300** and capable of driving the movable module **300** to retract through the through hole **110** into the housing **100** (that is, retracting into the inner chamber of the housing **100**) or to at least partially extend out of the housing **100**. In this case, the movable module **300** can be driven by the first driving mechanism **200** to at least partially extend out of the housing **100** to work. After the work is completed, the movable module **300** can be driven by the first driving mechanism **200** to retract into the housing **100** for accommodation in the housing **100**.

(11) The movable module **300** includes a first bracket **310**, a second driving mechanism **320**, a second bracket **330**, a functional module **340**, and a third driving mechanism **350**. The functional module **340** is rotatably disposed on the second bracket **330**, and the third driving mechanism **350** is disposed on the second bracket **330**, such that the second bracket **330** provides mounting positions for the third driving mechanism **350** and the functional module **340**. Specifically, the third driving mechanism **350** may be fastened on the second bracket **330** through welding, bonding, connection using connectors (for example, screw connectors), and the like. The functional module **340** may be disposed on the second bracket **330** through rotatable engagement in a shaft hole or bearing connection.

(12) The third driving mechanism **350** is connected to the functional module **340** and drives the functional module **340** to rotate around a first axis. In an optional implementation, the third driving mechanism **350** may be connected to the functional module **340** via a transmission mechanism. The third driving mechanism **350** can drive, via a first transmission mechanism, the functional module **340** to rotate, where the first transmission mechanism may be a chain transmission mechanism, a gear transmission mechanism, a belt transmission mechanism, or the like. The specific type of the first transmission mechanism is not limited in embodiments of this application.

(13) The second bracket **330** is rotatably disposed on the first bracket **310**, and the second driving mechanism **320** is disposed on the first bracket **310**, such that the first bracket **310** can provide mounting positions for the second driving mechanism **320** and the second bracket **330**. The second driving mechanism **320** is capable of driving the second bracket **330** to rotate around a second axis, and the functional module **340** is capable of rotating with the second bracket **330**, where the first axis and the second axis intersect or lie on different planes. In this case, the functional module **340** can rotate around the first axis, and can also rotate around the second axis, allowing for more flexible anti-shake rotation of the functional module **340**, thus implementing multi-directional anti-shake for the electronic device.

(14) Similarly, the second bracket **330** may also be rotatably mounted on the first bracket **310** through rotatable engagement in a shaft hole or through a bearing. The second driving mechanism **320** may be fastened on the first bracket **310** through welding, bonding, connection using connectors (for example, screw connectors), and the like.

(15) In this embodiment of this application, the functional module **340** includes an anti-shake micro-cloud gimbal **341** and a camera **342**, the camera **342** being movably disposed on the anti-

shake micro-cloud gimbal **341**. The camera **342** is a shooting function device of the functional module **340**, and the camera **342** is capable of rotating relative to the anti-shake micro-cloud gimbal **341**, which can also implement anti-shake. It should be noted that the anti-shake micro-cloud gimbal **341** is typically configured with an electromagnetic driving mechanism, and such electromagnetic driving mechanism can drive the camera **342** to rotate so as to implement anti-shake rotation of the camera **342**. This is a widely-known technology, and details are not described herein.

(16) Rotation of the functional module **340** around the first axis allows the camera **342** to rotate around the first axis, and rotation of the functional module **340** around the second axis allows the camera **342** to rotate around the second axis.

(17) During operation of the electronic device, when a user uses the functional module **340** of the electronic device, the first driving mechanism **200** can drive the movable module **300** to at least partially extend out of the housing **100** through the through hole **110**, so that the functional module **340** extends out of the housing **100** through the through hole **110**. In this case, the functional module **340** is not limited by the housing **100** and can be driven by the second driving mechanism **320** to rotate around the second axis and by the third driving mechanism **350** to rotate around the first axis, thereby implementing anti-shake rotation within a large angle range.

(18) In this embodiment of this application, the electronic device has a first anti-shake state and a second anti-shake state, where in a case that the electronic device is in the first anti-shake state, at least one of the second driving mechanism **320** and the third driving mechanism **350** drives the functional module **340** to rotate. Specifically, in a feasible driving process, the third driving mechanism **350** stops operating, and the second driving mechanism **320** can drive the second bracket **330** to rotate around the second axis, which in turn drives the functional module **340** to also rotate around the second axis, so that the camera **342** can rotate around the second axis, thus finally implementing anti-shake in one direction by the camera **342** rotating around the second axis.

(19) In another feasible driving process, the second driving mechanism **320** stops operating, and the third driving mechanism **350** drives the functional module **340** to rotate around the first axis, so that the camera **342** can rotate around the first axis, thus finally implementing anti-shake in another direction by the camera **342** rotating around the first axis.

(20) In still another feasible driving process, the second driving mechanism **320** and the third driving mechanism **350** operate simultaneously, so that the camera **342** also rotates around the second axis while rotating around the first axis, thus implementing more dimensional rotation of the camera **342** for anti-shake.

(21) In a case that the electronic device is in the second anti-shake state, the camera **342** is capable of rotating relative to the anti-shake micro-cloud gimbal **341**, thereby implementing anti-shake through rotation relative to the anti-shake micro-cloud gimbal **341**.

(22) It should be noted that in this embodiment of this application, a rotation angle range of the functional module **340** rotating around the first axis or the second axis is a first angle range, that is, a rotation angle range of the camera **342** rotating around the first axis or the second axis is the first angle range.

(23) A rotation angle range of the camera **342** rotating relative to the anti-shake micro-cloud gimbal **341** is a second angle range. The second angle range is smaller than the first angle range. For example, the first angle range of the camera **342** rotating around the first axis or the second axis may be 5°, and the second angle range of the camera **342** rotating relative to the anti-shake micro-cloud gimbal **341** may be 2°.

(24) The electronic device disclosed in embodiments of this application incorporates improvements to the related technology such that the camera **342** extends out of the housing **100** and is driven by the second driving mechanism **320** and/or the third driving mechanism **350** to implement anti-shake within a large angle range, and the camera **342** can also implement anti-shake within a small angle range relative to the anti-shake micro-cloud gimbal **341**. In this way, the electronic device

can select a corresponding anti-shake state according to a shaking angle during shooting, which can improve the anti-shake effects.

(25) In addition, in the electronic device disclosed in the embodiments of this application, during anti-shake rotation of the camera **342** around the first axis or the second axis, the camera **342** is not interfered with by other devices since it has extended out of the housing **100**. This allows for large-angle anti-shake and avoids the problem of poor anti-shake effects caused by a small anti-shake compensation angle.

(26) In an optional solution, the second axis may be an axis extending along a running direction of the through hole **110**, and the second axis may be perpendicular to the first axis. Optionally, the first axis and an optical axis of the camera **342** intersect or lie on different planes. Certainly, the first axis and the second axis may alternatively not be perpendicular to each other. For example, an included angle between the first axis and the second axis may be 30°, 45°, 60°, or the like.

(27) In this embodiment of this application, the functional module **340** may further include a fill light assembly **343**. The fill light assembly **343** may be fixedly connected to the anti-shake micro-cloud gimbal **341**, and a fill light direction of the fill light assembly **343** is consistent with a shooting direction of the camera **342**, so that the fill light assembly **343** can fill light for the camera **342**, thus improving clarity of images shot by the camera **342** in a dark environment, and ultimately further improving shooting effects of the electronic device.

(28) In this embodiment of this application, the third driving mechanism **350** and the fill light assembly **343** may be disposed on a first side of the second axis, and the anti-shake micro-cloud gimbal **341** and the camera **342** may be disposed on a second side of the second axis. Such layout can make structures and weights on two sides of the second axis more balanced, helping the functional module **340** rotate more stably around the second axis. In addition, such layout allows the functional module **340** to be made smaller in size along the second axis direction, which is conducive to reducing movement stroke of the movable module **300** driven by the first driving mechanism **200**, thus providing more sufficient mounting space for other components in the electronic device.

(29) In this embodiment of this application, the first driving mechanism **200** may be connected to the movable module **300** via a second transmission mechanism. Specifically, the second transmission mechanism may be a chain transmission mechanism, a gear transmission mechanism, a belt transmission mechanism, or the like. A specific type of the second transmission mechanism is not limited in this embodiment of this application.

(30) In a more preferable solution, a base **400** may be provided in the housing **100**, an elastic structural member **500** may be provided between the first bracket **310** and the base **400**, the elastic structural member **500** may be connected to a first side edge of the first bracket **310**, the first driving mechanism **200** may be connected to a second side edge of the first bracket **310**, and the first side edge and the second side edge are two side edges of the first bracket **310** that face away from each other. During operation of the first driving mechanism **200**, the first driving mechanism **200** drives the movable module **300** to move so as to make the elastic structural member **500** expand or contract, and the elastic structural member **500** serves as a buffer through its deformation to avoid great collision caused by excessively fast moving of the movable module **300**. For example, in a process of the first driving mechanism **200** driving the movable module **300** to retract into the housing **100**, the elastic structural member **500** may be compressed to deform so as to produce a buffering force opposite to the direction of retraction of the movable module **300**. This prevents the movable module **300** from abruptly retracting into the housing **100**, thus avoiding great collision and improving stability of retraction of the movable module **300**. In a process of the first driving mechanism **200** driving the movable module **300** to extend out of the housing **100**, the elastic structural member **500** is stretched, so as to prevent the movable module **300** from extending out of the housing **100** too fast.

(31) In addition, the elastic structural member **500** and the first driving mechanism **200** are

disposed at the first side edge and the second side edge respectively, which can make the force acting on the movable module **300** more balanced. This avoids the movable module **300** being subjected to large lateral force, facilitating more stable movement of the movable module **300**.

(32) In a further technical solution, the electronic device may further include a guide member **600**. Specifically, the base **400** is fixedly disposed in the inner chamber of the housing **100**, one end of the guide member **600** is fixedly connected to the base **400**, a guide direction of the guide member **600** may be consistent with a running direction of the through hole **110**, the other end of the guide member **600** is a free end, and the elastic structural member **500** is sleeved on the guide member **600**. In this way, the elastic structural member **500** deforms in the guide direction of the guide member **600**, allowing for more precise deformation of the elastic structural member **500**, thus improving buffering effects of the elastic structural member **500**.

(33) In a still further technical solution, the first side edge of the first bracket **310** may be provided with a sleeve member **700**. Specifically, the sleeve member **700** may be sleeved on the guide member **600**, and the elastic structural member **500** is elastically located between the sleeve member **700** and the base **400**. This allows the guide member **600** to be fixedly disposed between the base **400** and the first bracket **310**, while avoiding the elastic structural member **500** being fixedly connected to the first bracket **310**. This facilitates replacement of the elastic structural member **500** alone in case of damage to the elastic structural member **500**, avoiding the problem of high maintenance costs caused by the need for maintenance personnel to replace the elastic structural member **500** and the first bracket **310** together.

(34) In this embodiment of this application, a first circuit board **800** may be fastened in the housing **100**, a second circuit board **900** may be provided on the first bracket **310**, the second circuit board **900** is electrically connected to the first circuit board **800** via a first flexible electrical connector, and the first flexible electrical connector may be a flexible cable or a flexible circuit board. In an optional implementation, the guide member **600** may be a hollow guide rod. The first flexible electrical connector can pass through the hollow guide rod, thereby avoiding entanglement between the first flexible electrical connector and other components of the electronic device when the movable module **300** extends or retracts through the through hole **110**.

(35) Similarly, a rotating shaft **120** may be provided in the housing **100**, and the second bracket **330** is rotatably engaged with the first bracket **310** via the rotating shaft **120**. The rotating shaft **120** may be a hollow shaft with a central hole, and the third driving mechanism **350** and the functional module **340** may be electrically connected to the second circuit board **900** via a second flexible electrical connector that runs through the central hole. Optionally, the second flexible electrical connector may be a flexible circuit board or a flexible cable.

(36) Specifically, the third driving mechanism **350** may be electrically connected to the second circuit board **900** via the second flexible electrical connector, and the second flexible electrical connector runs through the central hole, avoiding entanglement between the second flexible electrical connector and other components of the electronic device during rotation of the second bracket **330**.

(37) In this embodiment of this application, the third driving mechanism **350** is capable of driving the functional module **340** to switch between a first angle and a second angle. In a case that the functional module **340** is at the first angle, the functional module **340** faces toward a first direction, so that the camera **342** can shoot images in the first direction and the fill light assembly **343** can face toward the first direction to provide light to the environment, thereby improving clarity of the images shot by the camera **342** in the first direction.

(38) In a case that the functional module **340** is at the second angle, the functional module **340** faces toward a second direction, so that the camera **342** can shoot images in the second direction and the fill light assembly **343** can provide light to the environment in the second direction, thereby improving clarity of the images shot by the camera **342** in the second direction. In addition, the first direction and the second direction may be opposite directions, thereby increasing the magnitude of

the camera **342** rotating around the first axis, and expanding the shooting range of the camera **342**. In this case, the functional module **340** enables the camera **342** to act as both a front-facing camera and a rear-facing camera, facilitating configuration of the electronic device with fewer cameras.

(39) The electronic device disclosed in this embodiment of this application may be a mobile phone, a tablet computer, an e-book reader, a wearable device (for example, smart glasses and a smart watch), a game console, a medical device, or the like. A specific type of the electronic device is not limited in this embodiment of this application.

(40) An embodiment of this application discloses a control method of electronic device, where the electronic device involved is the electronic device described in the foregoing embodiment, and the disclosed control method includes the following steps.

(41) Step **101**. Receive control information.

(42) In this step, the electronic device can measure, using a measuring device (such as a gyroscope), a shaking angle of the electronic device during shooting, so as to generate control information based on the shaking angle. This process may be completed by a processor of the electronic device. Specifically, the electronic device may generate different control information according to magnitude of the shaking angle. In a case of large shaking angle, the electronic device generates first control information; and in a case of small shaking angle, the electronic device generates second control information.

(43) Specifically, the magnitude of the shaking angle may be determined based on whether the shaking angle exceeds a preset threshold. For example, in a case that the shaking angle is larger than the preset threshold (for example, 4°), indicating a large shaking angle, the electronic device generates the first control information; and in a case that the shaking angle is less than the preset threshold, indicating a small shaking angle, the electronic device generates the second control information.

(44) Step **102**. In a case that the control information is first control information, control at least one of the second driving mechanism **320** and the third driving mechanism **350** to drive the functional module **340** to rotate.

(45) In this step, in a case of large shaking angle, the second driving mechanism **320** and the third driving mechanism **350** can drive the functional module **340** to rotate, allowing the camera **342** to implement large-angle anti-shake around the first axis or the second axis.

(46) Step **103**. In a case that the control information is second control information, control the camera **342** to rotate relative to the anti-shake micro-cloud gimbal **341**.

(47) In this step, in a case of small shaking angle, the camera **342** can implement small-angle anti-shake range relative to the anti-shake micro-cloud gimbal.

(48) In the process of a user shooting images with the camera **342**, the electronic device can make, according to a shaking state of the electronic device, at least one of the second driving mechanism **320** and the third driving mechanism **350** drive the functional module **340** to rotate, or the electronic device can make, according to a shaking state of the electronic device, the camera **342** rotate relative to the anti-shake micro-cloud gimbal **341**, thus implementing anti-shake within a large angle range and within a small angle range.

(49) The electronic device disclosed in the embodiments of this application is capable of implementing the processes implemented in the control method in the foregoing method embodiment. To avoid repetition, details are not described herein again.

(50) An embodiment of this application discloses a control apparatus of electronic device, where the electronic device involved is the electronic device described in the foregoing embodiment, and the disclosed control apparatus includes: a receiving module configured to receive control information; a first control module configured to, in a case that the control information is first control information, control at least one of the second driving mechanism **320** and the third driving mechanism **350** to drive the functional module **340** to rotate; and a second control module configured to, in a case that the control information is second control information, control the

camera **342** to rotate relative to the anti-shake micro-cloud gimbal **341**.

(51) Since the control apparatus disclosed in this embodiment of this application corresponds to the control method described above, for corresponding information, reference may be made to the foregoing description of the control method. Details are not described herein again.

(52) In the process of a user shooting images with the camera **342**, the receiving module can receive control information of the electronic device in a shaking state. If the control information is first control information, the first control module controls at least one of the second driving mechanism **320** and the third driving mechanism **350** to drive the functional module **340** to rotate, so that the electronic device implements anti-shake in the first anti-shake state. If the control information is second control information, the second control module controls the camera **342** to rotate relative to the anti-shake micro-cloud gimbal **341** for anti-shake, so that the electronic device implements anti-shake in the second anti-shake state.

(53) An embodiment of this application discloses a terminal device **1200** including a processor **1210**, a memory **1209**, and a program or instructions stored in the memory **1209** and capable of running on the processor **1210**. When the program or instructions are executed by the processor **1210**, the processes of any of the foregoing method embodiments are implemented, with the same technical effects achieved. To avoid repetition, details are not described herein again.

(54) FIG. 3 is a schematic diagram of a hardware structure of a terminal device **1200** for implementing the embodiments of this application.

(55) An embodiment of this application discloses a readable storage medium, where the readable storage medium stores a program or instructions, and when the program or instructions are executed by a processor, the processes of any of the foregoing method embodiments are implemented, with the same technical effects achieved. To avoid repetition, details are not described herein again. The readable storage medium may be a read-only memory (ROM), a random access memory (RAM), a magnetic disk, an optical disc, or the like.

(56) The terminal device **1200** includes but is not limited to components such as a radio frequency unit **1201**, a network module **1202**, an audio output unit **1203**, an input unit **1204**, a sensor **1205**, a display unit **1206**, a user input unit **1207**, an interface unit **1208**, a memory **1209**, a processor **1210**, and a power supply **1211**. Persons skilled in the art can understand that the structure of the terminal device **1200** shown in FIG. 3 does not constitute any limitation on the terminal device **1200**. The terminal device **1200** may include more or fewer components than shown in the figure, or combine some components, or arrange the components differently.

(57) The sensor **1205** is configured to measure a shaking angle of the electronic device during shooting, and the processor **1210** is configured to receive and differentiate control information. If the control information is first control information, the processor **1210** controls at least one of the second driving mechanism **320** and the third driving mechanism **350** to drive the functional module **340** to rotate; and if the control information is second control information, the processor **1210** controls the camera **342** to rotate relative to the anti-shake micro-cloud gimbal **341**, thereby avoiding shaking caused by the user's difficulty in maintaining a stationary shooting posture while shooting images with the electronic device. This achieves anti-shake compensation and ultimately improves quality of images shot by a shooting apparatus.

(58) It should be understood that in this embodiment of this application, the radio frequency unit **1201** may be configured to transmit and receive information, or to transmit and receive signals in a call process, and specifically, after receiving downlink data from a base station, transmit the downlink information to the processor **1210** for processing, and in addition, transmit uplink data to the base station. Generally, the radio frequency unit **1201** includes but is not limited to an antenna, at least one amplifier, a transceiver, a coupler, a low noise amplifier, and a duplexer. In addition, the radio frequency unit **1201** may further communicate with a network and other devices via a wireless communications system.

(59) The terminal device **1200** provides a user with wireless broadband internet access by using the

network module **1202**, for example, helping the user to transmit and receive e-mails, browse web pages, and access streaming media.

(60) The audio output unit **1203** may convert audio data received by the radio frequency unit **1201** or the network module **1202** or stored in the memory **1209** into an audio signal and output the audio signal as a sound. In addition, the audio output unit **1203** may further provide audio output (for example, a call signal received sound or a message received sound) related to a specific function performed by the terminal device **1200**. The audio output unit **1203** includes a speaker, a buzzer, a telephone receiver, and the like.

(61) The input unit **1204** is configured to receive an audio or video signal. The input unit **1204** may include a graphics processing unit (GPU) **12041** and a microphone **12042**. The graphics processing unit **12041** processes image data of a static picture or video that is obtained by an image capture apparatus (for example, a camera **342**) in an image or video capture mode. A processed image frame may be displayed on the display unit **1206**. An image frame processed by the graphics processing unit **12041** may be stored in the memory **1209** (or other storage media) or sent by the radio frequency unit **1201** or the network module **1202**. The microphone **12042** can receive a sound and can process the sound into audio data. Under a telephone call mode, the processed audio data can be converted into a format that can be sent as an output to a mobile communication base station by the radio frequency unit **1201**.

(62) The terminal device **1200** further includes at least one sensor **1205**, for example, an optical sensor, a motion sensor, and other sensors. Specifically, the optical sensor includes an ambient light sensor and a proximity sensor, where the ambient light sensor can adjust luminance of the display panel **12061** according to brightness of ambient light, and the proximity sensor can turn off the display panel **12061** and/or backlight when the terminal device **1200** is moved to an ear. As a type of motion sensor, an accelerometer sensor can detect magnitudes of accelerations in various directions (typically three axes), can detect the magnitude and direction of gravity when the mobile phone is in a static state, and can be used for posture recognition (such as screen switching between portrait and landscape, related games, and magnetometer posture calibration) of the terminal device **1200**, functions related to vibration recognition (such as pedometer and tapping), and the like. The sensor **1205** may also include a fingerprint sensor, a pressure sensor, an iris sensor, a molecular sensor, a gyroscope, a barometer, a hygrometer, a thermometer, an infrared sensor, and the like. Details are not described herein.

(63) The display unit **1206** is configured to display information input by the user or information provided to the user. The display unit **1206** may include a display panel **12061**, and the display panel **12061** may be configured in a form of a liquid crystal display (LCD), an organic light-emitting diode (OLED) display, or the like.

(64) The user input unit **1207** may be configured to receive input digit or character information and generate key signal input related to user setting and function control of the terminal device **1200**. Specifically, the user input unit **1207** includes a touch panel **12071** and other input devices **12072**. The touch panel **12071**, also referred to as a touchscreen, may capture a touch operation performed by a user on or near the touch panel **12071** (for example, an operation performed by the user on the touch panel **12071** or near the touch panel **12071** by using any appropriate object or accessory such as a finger or a stylus). The touch panel **12071** may include two parts: a touch detection apparatus and a touch controller. The touch detection apparatus detects a touch position of the user, detects a signal brought by a touch operation, and transmits the signal to the touch controller. The touch controller receives touch information from the touch detection apparatus, converts the touch information to point coordinates, and sends the point coordinates to the processor **1210**, and receives and executes a command sent by the processor **1210**. In addition, the touch panel **12071** may be implemented in a plurality of forms, for example, as a resistive, capacitive, infrared, or surface acoustic wave touch panel. The user input unit **1207** may further include other input devices **12072** in addition to the touch panel **12071**. Specifically, the other input devices **12072**

may include but are not limited to a physical keyboard, a function button (for example, a volume control button or a power on/off button), a trackball, a mouse, and a joystick. Details are not described herein.

(65) Further, the touch panel **12071** may cover the display panel **12061**. After detecting a touch operation on or near the touch panel **12071**, the touch panel **12071** transmits information about the touch operation to the processor **1210** for the processor **1210** to determine a touch event type, and then the processor **1210** provides a corresponding visual output on the display panel **12061** based on the touch event type. Although in FIG. 3, the touch panel **12071** and the display panel **12061** act as two separate parts to implement input and output functions of the terminal device **1200**, in some embodiments, the touch panel **12071** and the display panel **12061** may be integrated to implement the input and output functions of the terminal device **1200**. This is not specifically limited herein.

(66) The interface unit **1208** is an interface connecting an external apparatus to the terminal device **1200**. For example, the external apparatus may include a wired or wireless headphone port, an external power (or battery charger) port, a wired or wireless data port, a memory card port, a port for connecting an apparatus provided with a recognition module, an audio input/output (I/O) port, a video I/O port, an earphone port, and the like. The interface unit **1208** may be configured to receive input (for example, data information and power) from an external apparatus and transmit the received input to one or more elements in the terminal device **1200** or may be configured to transmit data between the terminal device **1200** and the external apparatus.

(67) The memory **1209** may be configured to store a software program and various data. The memory **1209** may mainly include a program storage area and a data storage area. The program storage area may store an operating system, an application program required for at least one function (for example, a sound play function and an image play function), and the like. The data storage area may store data created based on use of the electronic device (for example, audio data and a phone book), and the like. In addition, the memory **1209** may include a high-speed random access memory, or may further include a non-volatile memory, for example, at least one magnetic disk storage device, flash memory device, or another volatile solid-state storage device.

(68) The processor **1210** is a control center of the terminal device **1200**, and is connected to the components of the terminal device **1200** via various interfaces and lines. By running or executing software programs and/or modules stored in the memory **1209** and calling data stored in the memory **1209**, the processor **1210** executes various functions of the terminal device **1200** and processes data so as to perform overall monitoring on the terminal device **1200**. The processor **1210** may include one or more processing units. Optionally, the processor **1210** may integrate an application processor and a modem processor. The application processor mainly processes an operating system, a user interface, application programs, and the like. The modem processor mainly processes wireless communication. It can be understood that the modem processor may alternatively be not integrated in the processor **1210**.

(69) The terminal device **1200** may further include the power supply **1211** (for example, battery) supplying power to the components. Optionally, the power supply **1211** may be logically connected to the processor **1210** via a power management system, so that functions such as charge management, discharge management, and power consumption management are implemented by using the power management system.

(70) In addition, the terminal device **1200** includes some functional modules that are not shown, details of which are not described herein.

(71) The foregoing embodiments of this application focus on the differences between the embodiments. As long as the different features of improvement in the embodiments are not contradictory, they can be combined to form a more preferred embodiment. For brevity, details are not repeated herein.

(72) The foregoing descriptions are merely embodiments of this application and are not intended to limit this application. For persons skilled in the art, this application may have various changes and

variations. Any modification, equivalent replacement, improvement, or the like made without departing from the spirit and principle of this application shall fall within the protection scope of claims of this application.

Claims

1. An electronic device, comprising a housing, a first driving mechanism, and a movable module, wherein the housing has an inner chamber and a through hole communicating with the inner chamber, the first driving mechanism is disposed in the inner chamber, and the first driving mechanism is connected to the movable module and capable of driving the movable module to retract through the through hole into the housing or at least partially extend out of the housing; wherein the movable module comprises a first bracket, a second driving mechanism, a second bracket, a functional module, and a third driving mechanism, wherein the functional module is rotatably disposed on the second bracket, the third driving mechanism is disposed on the second bracket, the third driving mechanism is connected to the functional module and drives the functional module to rotate around a first axis, the second bracket is rotatably disposed on the first bracket, the second driving mechanism is disposed on the first bracket, the second driving mechanism drives the second bracket to rotate around a second axis, the functional module is capable of rotating with the second bracket, and the first axis and the second axis intersect or lie on different planes; the functional module comprises an anti-shake micro-cloud gimbal and a camera, the camera being movably disposed on the anti-shake micro-cloud gimbal; and the electronic device has a first anti-shake state and a second anti-shake state, wherein in a case that the electronic device is in the first anti-shake state, at least one of the second driving mechanism or the third driving mechanism drives the functional module to rotate; and in a case that the electronic device is in the second anti-shake state, the camera is capable of rotating relative to the anti-shake micro-cloud gimbal; wherein a base is provided in the housing, and an elastic structural member is provided between the first bracket and the base, wherein the elastic structural member is connected to a first side edge of the first bracket, the first driving mechanism is connected to a second side edge of the first bracket, and the first side edge and the second side edge are two side edges of the first bracket that face away from each other; wherein the electronic device further comprises a guide member, wherein one end of the guide member is fixedly connected to the base, and the other end of the guide member is a free end; and the elastic structural member is sleeved on the guide member.

2. The electronic device according to claim 1, wherein the second axis is an axis extending along a running direction of the through hole, and the second axis is perpendicular to the first axis.

3. The electronic device according to claim 1, wherein the functional module further comprises a fill light assembly, wherein the fill light assembly is fixedly connected to the anti-shake micro-cloud gimbal, and a fill light direction of the fill light assembly is consistent with a shooting direction of the camera.

4. The electronic device according to claim 3, wherein the third driving mechanism and the fill light assembly are disposed on a first side of the second axis, and the anti-shake micro-cloud gimbal and the camera are disposed on a second side of the second axis.

5. The electronic device according to claim 1, wherein the first side edge of the first bracket is provided with a sleeve member, wherein the sleeve member is sleeved on the guide member, and the elastic structural member is elastically located between the sleeve member and the base.

6. The electronic device according to claim 1, wherein a first circuit board is fastened in the housing, a second circuit board is provided on the first bracket, the second circuit board is electrically connected to the first circuit board via a first flexible electrical connector, the second bracket is rotatably engaged with the first bracket via a rotating shaft, the rotating shaft is a hollow shaft with a central hole, and the third driving mechanism and the functional module are

electrically connected to the second circuit board via a second flexible electrical connector that runs through the central hole.

7. The electronic device according to claim 1, wherein the third driving mechanism is capable of driving the functional module to switch between a first angle and a second angle; and in a case that the functional module is at the first angle, the functional module faces toward a first direction, and in a case that the functional module is at the second angle, the functional module faces toward a second direction; wherein the first direction and the second direction are opposite directions.

8. A control method of an electronic device, wherein the electronic device is the electronic device according to claim 1, and the control method comprises: receiving control information; in a case that the control information is first control information, controlling at least one of the second driving mechanism or the third driving mechanism to drive the functional module to rotate; and in a case that the control information is second control information, controlling the camera to rotate relative to the anti-shake micro-cloud gimbal.

9. The method according to claim 8, wherein the second axis is an axis extending along a running direction of the through hole, and the second axis is perpendicular to the first axis.

10. The method according to claim 8, wherein the functional module further comprises a fill light assembly, wherein the fill light assembly is fixedly connected to the anti-shake micro-cloud gimbal, and a fill light direction of the fill light assembly is consistent with a shooting direction of the camera.

11. The method according to claim 10, wherein the third driving mechanism and the fill light assembly are disposed on a first side of the second axis, and the anti-shake micro-cloud gimbal and the camera are disposed on a second side of the second axis.

12. The method according to claim 8, wherein the first side edge of the first bracket is provided with a sleeve member, wherein the sleeve member is sleeved on the guide member, and the elastic structural member is elastically located between the sleeve member and the base.

13. The method according to claim 8, wherein a first circuit board is fastened in the housing, a second circuit board is provided on the first bracket, the second circuit board is electrically connected to the first circuit board via a first flexible electrical connector, the second bracket is rotatably engaged with the first bracket via a rotating shaft, the rotating shaft is a hollow shaft with a central hole, and the third driving mechanism and the functional module are electrically connected to the second circuit board via a second flexible electrical connector that runs through the central hole.

14. The method according to claim 8, wherein the third driving mechanism is capable of driving the functional module to switch between a first angle and a second angle; and in a case that the functional module is at the first angle, the functional module faces toward a first direction, and in a case that the functional module is at the second angle, the functional module faces toward a second direction; wherein the first direction and the second direction are opposite directions.

15. A terminal device, comprising a processor, a memory, and a program or instructions stored in the memory and capable of running on the processor, wherein when the program or instructions are executed by the processor, the steps of the control method according to claim 8 are implemented.

16. A non-transitory readable storage medium, wherein the readable storage medium stores a program or instructions, and when the program or instructions are executed by a processor, the steps of the control method according to claim 8 are implemented.
